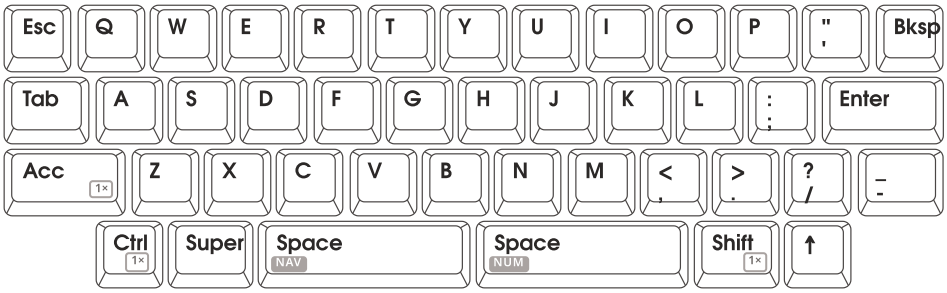
# Agar mini

## Custom ZMK Layout
## Key Function Manual

ZMK FIRMWARE

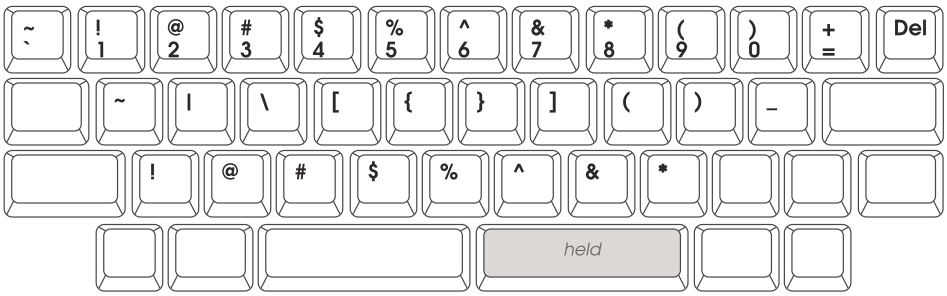
Quick tip: a **grey tag** is what the key does when held. **1×** marks a sticky key: tap it, then the next key. Blank keys pass through to Layer 0.

## Layer 0 BASE



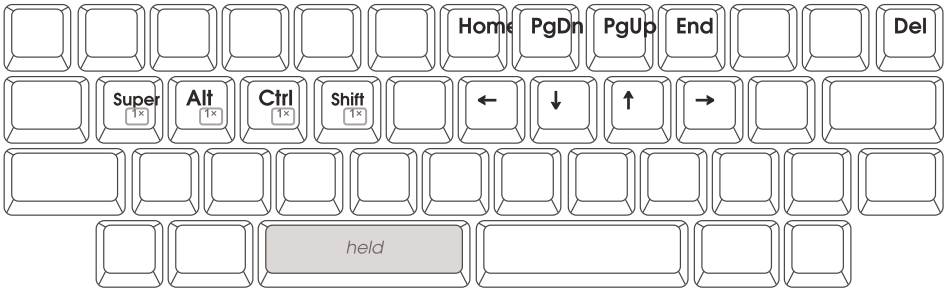

- 1× **Sticky:** tap, then the next key
- Acc **Accent key** (back of the left pinky): tap it, then the vowel, Layer 4
- NUM **Hold for Layer 1**, right Space
- NAV **Hold for Layer 2**, left Space
- **Both Spaces** for Layer 3
- ' **Plain apostrophe**; Shift gives "
- Super **Plain key**, hold it
- Shift **The only Shift**, right thumb
- ↑ **Up arrow** for the last command

## Layer 1 NUM



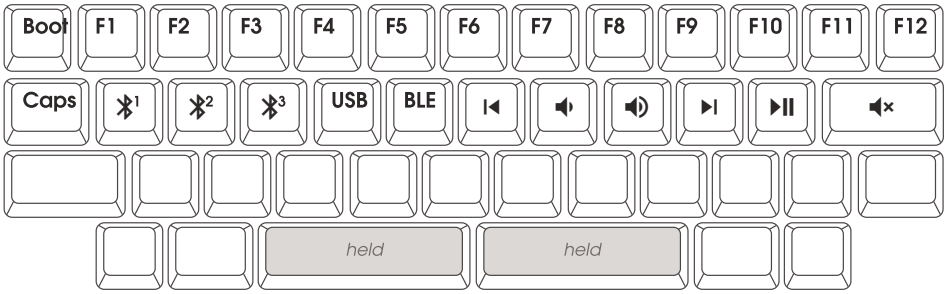

- **Hold right Space**
- 1–0 **Super + number** switches mango workspace
- . / - **Pass through** for `~/`, `1.5`, IPs
- ` ~ ^ **Plain characters**, not dead keys

## Layer 2 NAV



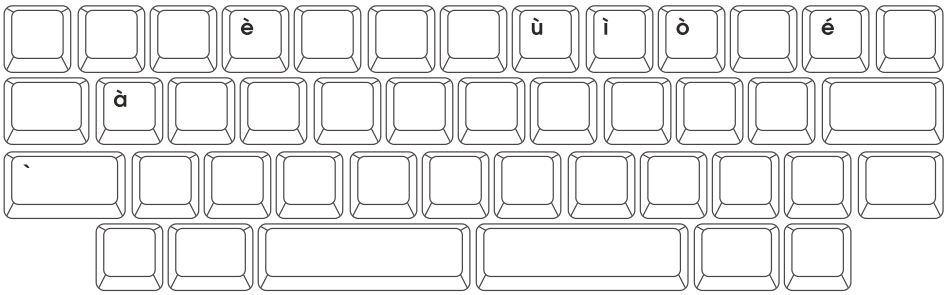

- **Hold left Space**
- ← ↓ ↑ → **Arrows on hjkl**
- 1× **Sticky modifiers:** tap `D`, tap `L` = Ctrl+→
- Home **Home, PgDn, PgUp, End** sit above the arrows

## Layer 3 FN




- **Hold both Spaces**
- BT1 **Bluetooth profile 1–3**; hold 3 s to clear its pairing
- USB **Switch output** to USB or BLE
- Boot **Bootloader:** hold 3 s; a tap restarts
- ⏮ ⏯ ⏭ **Media** previous, play/pause, next
- 🔉 🔊 🔇 **Volume** down, up, mute

## Layer 4 ACC




- 1× **Tap Acc**, then the key; expires after 1.5 s
- È **Capital:** tap Shift too, before or after Acc
- ` **Acc twice** types a backtick
- . **Other keys pass through**, so a stray Acc does no harm

# Everyday actions

| ACTION | SENDS | KEYS |
|---|---|---|
| è à ì ò ù | | tap Acc, tap the vowel |
| é | | tap Acc, tap ' |
| È | | tap Shift, tap Acc, tap E |
| Backtick | ` | tap Acc twice / hold R Space + Esc |
| Capital letter | | tap Shift (right thumb), tap the letter |
| Apostrophe / quote | ' / " | tap ' / tap Shift, tap ' |
| Newline in Claude Code | Shift+Enter | tap Shift, tap Enter |
| Question / exclamation mark | ? / ! | tap Shift, tap / / hold R Space + Z |
| Number | 1 … 0 | hold R Space + Q ... P |
| Backspace | | top-right corner, as on the Q4 |
| Run the last command again | Up, Enter | tap ↑ (bottom right), tap Enter |
| Copy / paste | Ctrl+C / Ctrl+V | tap Ctrl, tap C / V |
| Kitty copy / paste | Ctrl+Shift+C / V | tap Ctrl, tap Shift, tap C / V |
| Switch workspace | Super+3 | hold Super + hold R Space + E |
| Move window to workspace | Super+Shift+3 | tap Shift, then hold Super + hold R Space + E |
| Other mango shortcuts | Super+Return, Super+q | hold Super + the key |
| Focus direction | Super+Left | hold L Space, tap A, tap H |
| Kitty tab | Alt+3 | hold L Space, tap S (Alt), then hold R Space + E |
| Accept fish autosuggestion | Right | hold L Space + L |

No mango config changes are needed: every action sends exactly what a full-size keyboard would.

# Timing

| KEY | FLAVOR | TERM | QUICK-TAP | PRIOR IDLE |
|---|---|---|---|---|
| Space / NAV | balanced | 200 | 175 | 150 |
| Space / NUM | balanced | 200 | 175 | — |

| STICKY KEYS | RELEASE AFTER | QUICK RELEASE |
|---|---|---|
| Shift, Ctrl, Alt, Super | 1500 | yes |
| Acc (sticky layer) | 1500 | no |

- All values in milliseconds. Letters and ' have no hold behaviour: they type on press.
- Sticky keys apply to the next key only and expire after 1.5 s. Tapping Ctrl then Shift stacks them. They can also be held normally. Acc ignores modifiers, so Shift tapped between Acc and the vowel still gives the capital.
- Space / NUM has no prior-idle setting; otherwise typing `port 8080` quickly would give letters instead of digits.

## Design

1. **Thumbs hold, nails tap one key at a time.** No chords; ring fingers only reach Esc and Backspace, the pinky only Acc, with the back of the finger.
2. **Accents as two taps:** Acc, then the vowel, like ` + vowel on `us intl`.
3. **Sticky Shift, Ctrl and Alt:** tap, then the next key.
4. **' and " are always plain**, despite `us intl` dead keys.
5. **Super under the left thumb, sticky Shift under the right.** Backspace stays top-right.

## Requires `us intl`

Accents and plain quotes are typed as `us intl` dead-key sequences: ` + vowel for è, ' + e for é, ' + Space for '. On a device with a plain US layout, accents come out as `e and quotes get an extra space.

## Dropped from stock

Mouse keys (trackpad available), Win+V, Print Screen (mango uses Super+Shift+S), Insert, Pause, Scroll Lock, right Alt, arrow cluster (↑ kept bottom right, the rest on NAV).